30-DAY GIT LEARNING SERIES

GIT QUEST

LEVEL 11 | THE BIG IDEA

What is a Branch?

Week 2: Undo and Branch | Absolute Beginner

QUEST PROGRESS: DAY 11 OF 30

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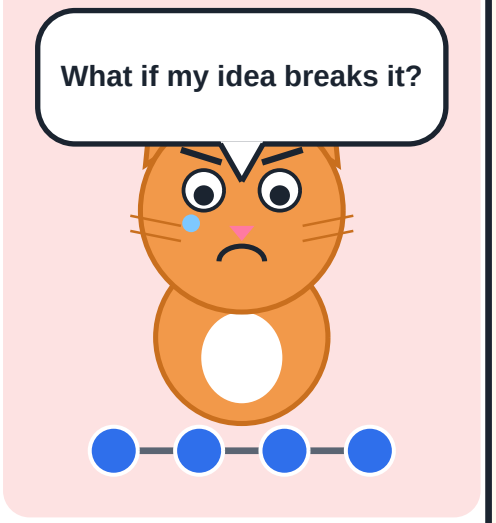
The Story: The Parallel Timeline

Commit the Cat wants to try a wild idea without breaking anything.

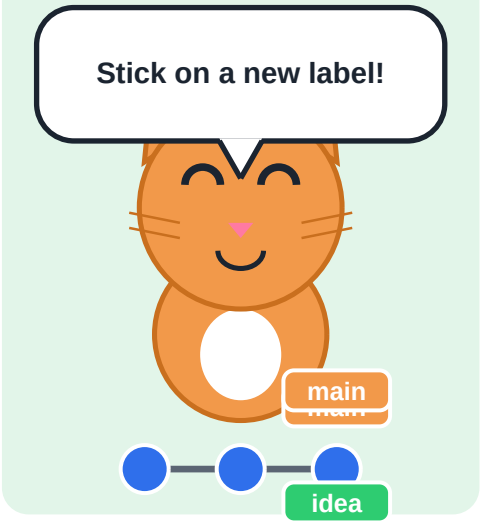
1



One timeline

So far every commit follows the last one. A risky idea could spoil the working version.

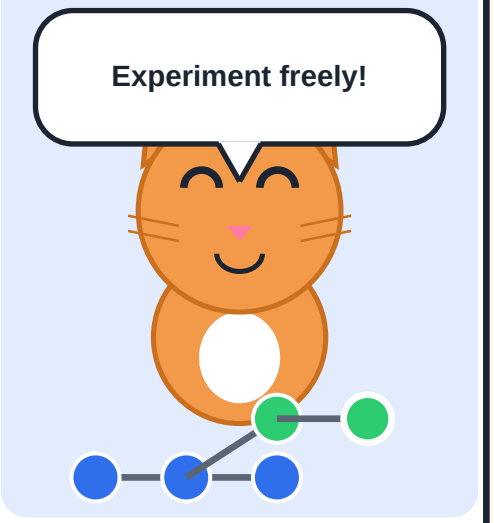
2



A sticky note

A branch is just a movable label pointing at a commit. Not a copy of your files.

3



A safe fork

Commits on the new branch leave main untouched. Main keeps its own timeline.

4

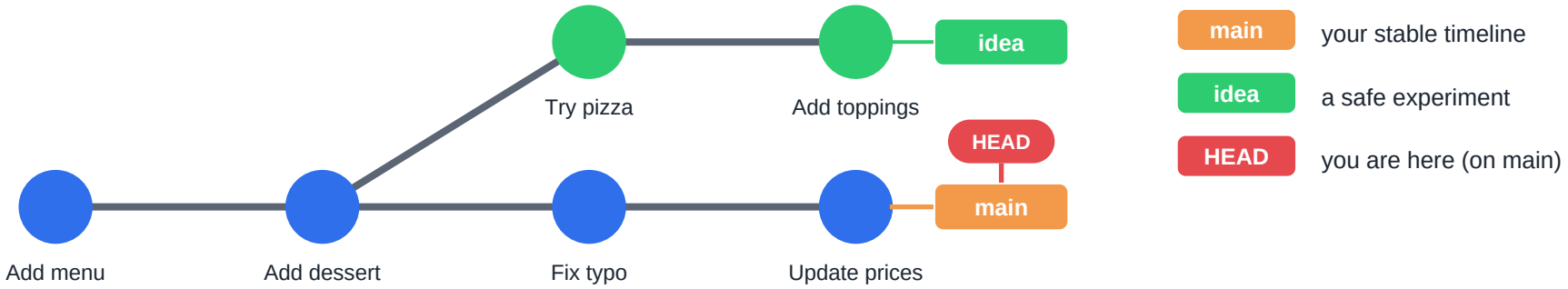
Merge

Happy with the experiment? Combine it back into main with a merge (Day 13).

Anatomy of a Branch

A branch is a sticky note on a commit. That's all.

PARALLEL TIMELINES



BRANCH

A movable label on one commit. Not a copy of your files.

main

The default branch: your first sticky note, made by git init -b main.

HEAD

The 'you are here' pin. It points at the branch you are on.

WHY USE BRANCHES?

- Try ideas safely
- Work in parallel
- Keep main stable
- Collaborate

Branches are cheap and fast. Make as many as you like!

Quest Board: Meet Your First Branch

Finish 4 quests, then collect today's commit.

1 Make a tiny history

```
$ git init -b main branch-lab && cd branch-lab
$ echo "Pasta" > menu.txt && git add menu.txt
$ git commit -m "Add menu"
```

2 Add a second commit

```
$ echo "Dessert" >> menu.txt && git add menu.txt
$ git commit -m "Add dessert"
```

3 See your only branch

```
$ git branch
$ git log --oneline --decorate
```

** main is the only branch. HEAD -> main means you are here.*

4 Stick on a second label

```
$ git branch idea
$ git log --oneline --decorate
$ git branch -d idea
```

(HEAD -> main, idea): two labels, one commit. Tomorrow you switch!

LOOT: TODAY'S COMMIT

COMMIT

days-1-to-11-summary.md

```
docs(day11): add days 1-11 summary and branch notes
```

Recap of Days 1 to 11 plus branch notes.

KEY WORDS

branch	a movable label on a commit
main	the default branch
HEAD	the 'you are here' pin

COMING TOMORROW (DAY 12)

```
$ git switch idea
```

Jump onto a branch and commit there.

Boss Fight and Bonus Loot

Cheat sheet, key words, quiz and your next unlock.

BONUS 1: DAY 11 CHEAT SHEET

```
git branch                      List branches (* = current)
git branch <name>               Make a new label, stay put
git branch -d <name>            Delete a branch label
git log --oneline --decorate    Show labels on commits
git log --oneline --graph --all Draw every branch
```

BONUS 2: THE PICTURE IN WORDS

```
branch  = movable label on one commit
main    = the default branch
HEAD    = you are here (points at a branch)
```

BONUS 3: DIAGRAM GITHUB DRAWS FOR YOU

```
```mermaid
gitGraph
  commit id: "Add menu"
  branch idea
  checkout idea
  commit id: "Try pizza"
```
```

BOSS FIGHT: QUICK QUIZ

1. What is a branch, really?
2. What does HEAD -> main mean?
3. Give two reasons to use branches.

Answers: 1) A movable label on a commit 2) You are on main 3) Try ideas safely; keep main stable

CHALLENGE

Draw (on paper) a history with a 'dark-mode' branch that has two commits while main gets one. Label HEAD.

LEVEL 12: git switch and git branch

Unlocks tomorrow. Jump between branches and commit on them.

LOCKED